

- Be careful when buying used IoT devices, as they could have been tampered with. It is better to consult a genuine authority and confirm the device's validation or buy from a certified authority.
- Research the vendor's device security measures as well as the features that they support.
- Modify the privacy and security settings of the device to meet your needs immediately after buying the device.
- It is better to disable features that are not used frequently.
- Install updates regularly, when they become available. It is a best practice to use the latest firmware updates.
- Ensure that an outage due to jamming or a network failure does not result in an insecure state of the installation.
- Verify if the smart features are required or if a normal device suffices for the purpose.

Best practices for designers of IoT frameworks and device manufacturers

- Always use SSL/TLS-encrypted connections for communication purposes.
- Check the SSL certificate and the certificate revocation list.
- Allow and encourage the use of strong passwords and change default passwords immediately.
- Provide a simple and secure update process with a chain of trust.
- Provide a standalone option that works without Internet and cloud connections.
- Prevent brute-force attacks at the login stage through account lockout measures or with multi-factor types of authentication mechanisms.
- Implement a smart fail-safe mechanism when the connection or power is lost or jammed.
- Remove unused tools and allow only trusted applications and software.
- Where applicable, security analytics

features should be provided in the device management strategy.

IoT developers and designers should include security at the start of the device development process, irrespective of whether the device is for the consumer market, the enterprise or industry. Incorporating security at the design phase always helps. Enabling security by default is very critical, as is providing the most recent operating systems and using secure hardware with the latest firmware versions.

Enabling PKI and digital certificates

Public key infrastructure (PKI) and 509 digital certificates play important and critical roles in the development of secure IoT devices. It is always a best practice to provide the trust and control needed to distribute and identify public encryption keys, secure data exchanges over networks and verify the identity.

API (application performance indicator) security

For any IoT environment, API security is essential to protect the integrity of data. As this data is being sent from IoT devices to back-end systems, we always have to make sure only authorised devices, developers and apps communicate with these APIs.

Patch management/continuous software updates

This is one crucial aspect in IoT security management. Providing

the means of updating devices and software either over network connections or through automation is critical. Having a coordinated disclosure of vulnerabilities is also important to updating devices as soon as possible. Consider end-of-life strategies as well.

Always remember that hard coded credentials should never be used nor be part of the design process. If there are any default credentials, users should immediately update them using strong passwords as described earlier, or follow multi-factor or biometric authentication mechanisms.

Hardware security

It is absolutely essential to make devices tamper-proof or tamper-evident, and this can be achieved by endpoint hardening.

Strong encryption is critical to securing communication between devices. It is always a best practice to encrypt data at rest and in transit using cryptographic algorithms.

IoT and operational system security are new to many security teams. It is critical to keep security staff up to date with new or unknown systems, enabling them to learn new architectures and programming languages to be ready for new security challenges. C-level and cyber security teams should receive regular training to keep up with modern threats and security measures. **END** 

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